

CALEB KUSUMO

MECHANICAL ENGINEERING / SYSTEMS DESIGN / BOSTON, MA

A complete, linear edition of the portfolio at kusumo.design



ENGINEERING PROFILE

Caleb Kusumo is a second-year Mechanical Engineering student at Boston University interested in the intersection of mechatronics, industrial design, and manufacturing systems. His work spans robotics, electronics, mechanical systems, camera repair, and hands-on manufacturing. He is the founder of CalebColor Film, a former FIRST Robotics Team 6560 captain and manufacturing lead, and a steering lead with Terrier Motorsport Formula SAE.

PROJECT ARCHIVE / 09 WORKS

MECHANICAL DESIGN | MECHATRONICS | OPTOMECHANICAL SYSTEMS



2025 Robot [REF_6560-25](#)

MECHATRONICS

Reefscape robot with vision tracking and a specialized intake for accurate coral manipulation.

SolidWorks / CNC / CAN Bus / Laser Cutting



2024 Robot [REF_6560-24](#)

MECHATRONICS

Crescendo robot with a high-speed swerve drivetrain and vision-aligned note delivery.

SolidWorks / Swerve / CNC



2023 Robot [REF_6560-23](#)

MECHATRONICS

Charged Up robot with a piston-actuated arm and gripper for capturing conical and square game pieces.

Java / SolidWorks / Pneumatics / CNC



RB67 Darkslide [REF_CK-01](#)

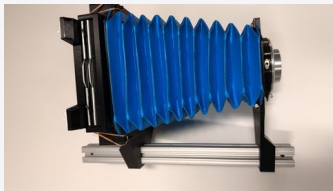
MECHANICAL DESIGN

High-precision linear-motion assembly designed around specialized low-friction coatings.

SolidWorks / Waterjet / 3D Printing

PROJECT ARCHIVE / CONTINUED

MECHANICAL DESIGN | MECHATRONICS | OPTOMECHANICAL SYSTEMS



4x5 Rail Camera [REF_CK-04](#)

OPTOMECHANICAL DESIGN

Modular large-format camera with four-axis movements and high-extension bellows.

SolidWorks / 3D Printing / Camera / Laser Cutting



Lutron Lighting [REF_CK-05](#)

ARCHITECTURAL SYSTEMS

A 1:100-scale gallery lighting display, placed third among 40 competition entries.

Fusion 360 / Circuits / Laser Cutting / 3D Printing / Waterjet



Waist Level Finder [REF_CK-06](#)

OPTOMECHANICAL DESIGN

Custom EF-mount, 1:1 ground-glass finder for through-the-lens waist-level viewing.

3D Printing / Optics / SolidWorks



Mamiya Lens Hood [REF_CK-07](#)

INDUSTRIAL DESIGN

Specialized printed lens hood for the Mamiya C330 TLR and its 80mm f/2.8 lens.

3D Printing / Camera / CAD



Camera Repair [REF_CK-08](#)

MECHANICAL SERVICE

Shutter timing, film transport, light seals, and lens service for film and digital cameras.

Calibration / Repair / Optics / Camera

EXPERIENCE & BACKGROUND

EDUCATION / ENGINEERING / ENTREPRENEURSHIP / SERVICE

EDUCATION

B.S. Mechanical Engineering - Dean's List (Fall 2025) | 2025-PRESENT

Boston University | Boston, MA

High School Diploma - Magna Cum Laude; Outstanding Engineer of the Year | 2021-2025

University High School | Irvine, CA

GPA 3.9 UW / 4.19 W

ENGINEERING EXPERIENCE

Steering Lead / Aero & Bodywork Team Member | SEP 2025-PRESENT

Terrier Motorsport Formula SAE | Boston, MA

Managing steering assembly design and fabrication while collaborating on aerodynamic components and bodywork integration. Applying composite and manufacturing techniques to improve vehicle performance.

Team Captain / Manufacturing & Assembly Lead | SEP 2022-APR 2025

FIRST Robotics Competition Team 6560 - Charging Champions | Irvine, CA

Led a 15+ person fabrication group and helped a 30+ member team move from design to a competition robot in under four weeks. Served as technician and drive-team member for REEFSCAPE and CRESCENDO; earned regional recognition and awards in control, autonomous, and creativity.

Mechanical Team Member | 2024-2025

Hydrogen Grand Prix | Irvine, CA

Assembled mechanical systems for a hydrogen-powered RC racer, focusing on endurance and fuel efficiency.

PROJECTS & ENTREPRENEURSHIP

Individual Competitor - Third Place | FEB 2026

Lutron Lighting Competition

Designed, built, wired, and presented a 1:100-scale gallery lighting solution judged on creativity, engineering, and aesthetics.

Founder / Lead Technician | 2023-PRESENT

CalebColor Film

Built a film sales and processing service, developing 200+ rolls and selling 300+ rolls. Service work includes shutter timing, film advance, screens, sensors, and light seals.

Freelance Photographer | 2022-PRESENT

Photography

Completed 100+ portrait, event, and wedding sessions. Led photography at community events and managed post-production workflows for K4U Dance Club and Replay Performing Arts.

LEADERSHIP & SERVICE

Mentor - Electrical Systems, CNC, Milling & Lathe | 2025-PRESENT

FIRST Robotics Competition Team 6560

Mentors junior members on electrical systems, CNC operation, milling, and lathe work for precise assembly.

Structures Lead | 2021-2025

Student Council

Led a 20+ member team building themed Spirit Night installations, including a life-sized Mystery Machine, Las Vegas sign, and gingerbread house.

TECHNICAL SKILLS

DESIGN & SOFTWARE

SolidWorks, Fusion 360, Adobe Lightroom, Photoshop, Illustrator

MANUFACTURING

CNC machining, lathe and milling, waterjet, laser cutting, SLA/FDM printing, TIG welding, vacuum forming, vinyl cutting

ELECTRONICS

Soldering, Arduino, Raspberry Pi, oscilloscope, general circuitry

DEVELOPMENT

Python, C++, Java, Next.js

CONTACT / AVAILABLE FOR TECHNICAL CONSULTATION

calebk@bu.edu | LinkedIn | GitHub

kusumo.design / Photography: calebcolor.com